

Charotar University of Science & Technology

M.Tech(CAD CAM) University Exam

ME 709 ADVANCE ENGINEERING OPTIMISATION TECHNIQUE

Year/Semester: 2nd sem/ 201 1

Total Marks: 70

Date:18/5/2011

Time: 10:00a.m to 01:00 p.m

- Instructions:* (i) Attempt **all** questions.
(ii) Figures to the right indicate **full** marks.
(iii) Make suitable assumptions and draw neat figures wherever required.
(iv) Write each section in separate answerbook.

SECTION 1

- Q-1** Five strip-layout selection attributes viz. economical material utilization (Ur), die cost (Dc), stamping operational cost (Oc), required production rate (Pr) and job accuracy (Ja) are presented in Table 1 and the estimated quantitative values are assigned to Ur, Dc, Oc, Pr and qualitative values assigned to Ja .Find the optimum layout using Modified TOPSIS method. **10**

Table 1: Strip layout attributes

Layout	Ur	Dc	Oc	Pr	Ja
(a)	0.26	25000	130	80	4
(b)	0.40	28560	138	120	3
(c)	0.33	31109	90	150	3
(d)	0.32	31702	150	125	2
(e)	0.31	32390	160	110	2
(f)	0.31	32663	116	108	2

Ur is expressed in %; Dc in Rupees; Oc in Rupees/1000 pieces; Pr in pieces/minute.
The normalized weights of each attribute are: WUr = 0.2815; WDc = 0.1054; WOc = 0.0501; WPr = 0.2815; and WJa = 0.2815.

Q.2

- a.** Discuss the algorithm of Hooke and Jeeves' Method. **5**
- b.** State Kuln Tucker's conditions. For the following problem check at point (0,0) whether constraint qualification and Kuln Tucker conditions are satisfied or not. **10**
- Minimise $f(x_1, x_2) = (x_1 + 1)^2 - x_2$
subjected to
 $g_1(x_1, x_2) = x_1^2 - 2x_2 \leq 0$
 $g_2(x_1, x_2) = x_1^2 + 2x_2 \leq 0$
- OR**
- b.** 1. Define a convex function. Determine whether $f(x)=e^x$ is convex or not **3**
2. Find extreme points of the function **4**
- $f(x_1, x_2) = x_1^2 + x_2^2 + 2x_1^2 + 4x_2^2 - 4$
3. Define and explain saddle point with example. **3**

Q.3 Write short note on the following(Any two)

10

1. Tabu search algorithm
2. Particle swarm algorithm
3. Ant colony optimisation

SECTION 2

Q.4

a. Solve the following using Fibonacci Method in the interval (0, 3) for n=7.

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$$f(x) = \frac{0.5}{\sqrt{1+x^2}} - \sqrt{(1+x^2) \left(1 - \frac{0.5}{\sqrt{1+x^2}}\right)}$$

Q-5 Attempt any three.

a. Find the value of x in the interval (0,1) which minimizes the function

6

$$f(x) = x(x - 1.5)$$

to within ± 0.05 by golden section method.

b. Find the no. of experiments to be conducted in the following methods to obtain value of $L_n/L_0=0.001$.

6

- 1.Interval halving method
- 2.Fibonacci method
- 3.Golden section method

c. Write C programme for Fibonacci method for finding minimum value of any function.

6

d. Explain Interval Halving method and dichotomous search method..

6

e. Define unimodal functions and explain its properties.

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Q.6 Find the minimum of the function

7

$$f(x) = 0.65 - \frac{0.75}{1+x^2} - 0.65x \tan^{-1} \frac{1}{x}$$

Using the Newton Raphson method with the starting point $x_1=0.1$. Use $\epsilon=0.01$ for checking the convergence.

OR

The efficiency of screw jack is given by

$$\eta = \frac{\tan \alpha}{\tan(\alpha + \phi)}$$

where α is the lead angle and ϕ is a constant. Prove that the efficiency of the screw jack will be maximum when $\alpha=45^\circ - \phi/2$ with

$$\eta_{\max} = \frac{1 - \sin \phi}{1 + \sin \phi}$$